



Advantech SOP - Implement RAG on Jetson Device

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What is RAG?

Retrieval-Augmented Generation (RAG) is an **AI technique** that combines information retrieval with text generation to improve the accuracy and relevance of responses. It works by:

1. **Retrieving** relevant documents from a knowledge base (e.g., a database, a vector store, or an external source) based on the user's query.
2. **Augmenting** the input by providing the retrieved documents to a language model.
3. **Generating** a final response using both the retrieved data and the model's internal knowledge.

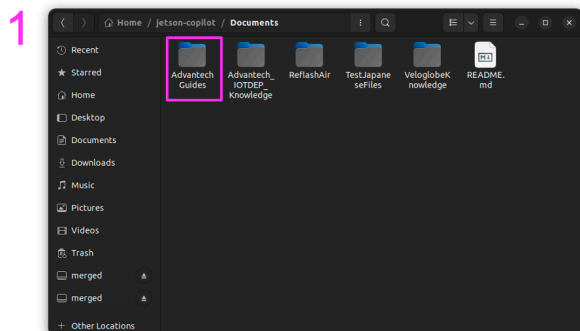
With RAG, you **don't train a model from scratch**. Instead, you create a database (knowledge base) with your documents, and an existing LLM (like GPT) retrieves the relevant info from it in real-time.

For this we used an open source repository from **NVIDIA** called **Jetson Copilot**

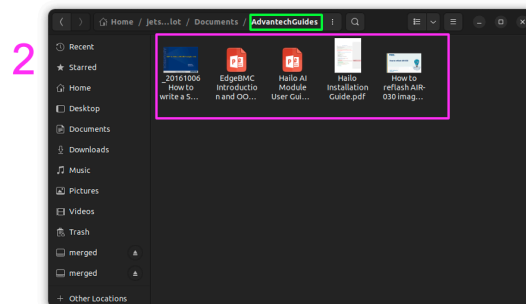
Install jetson-containers

1. Follow this documentation to install the application on your Jetson device:
https://www.jetson-ai-lab.com/tutorial_jetson-copilot.html
2. Look at the video's in this tutorial for better understanding on how to use it.
3. Down here we explained it a bit further with images.

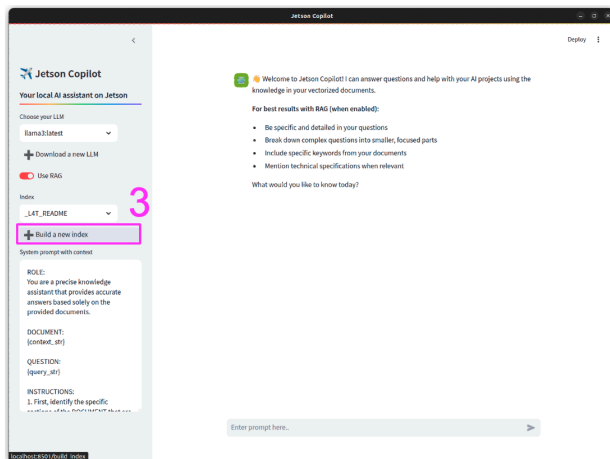
Explained with images



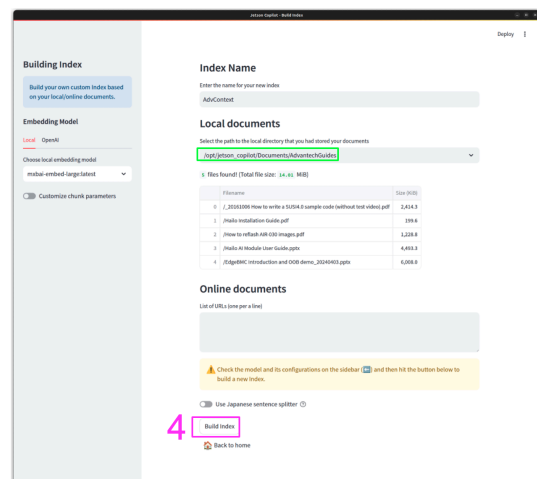
Create a knowledge base creating a new folder inside jetson-copilot/Document



Add your files to your created folder
Supported files: 'txt', 'csv', 'docx', 'epub', 'hwp', 'ipynb', 'jpeg', 'jpg', 'mbox', 'md', 'mp3', 'mp4', 'pdf', 'png', 'ppt', 'pptm', 'pptx', 'xlsx', 'xls'



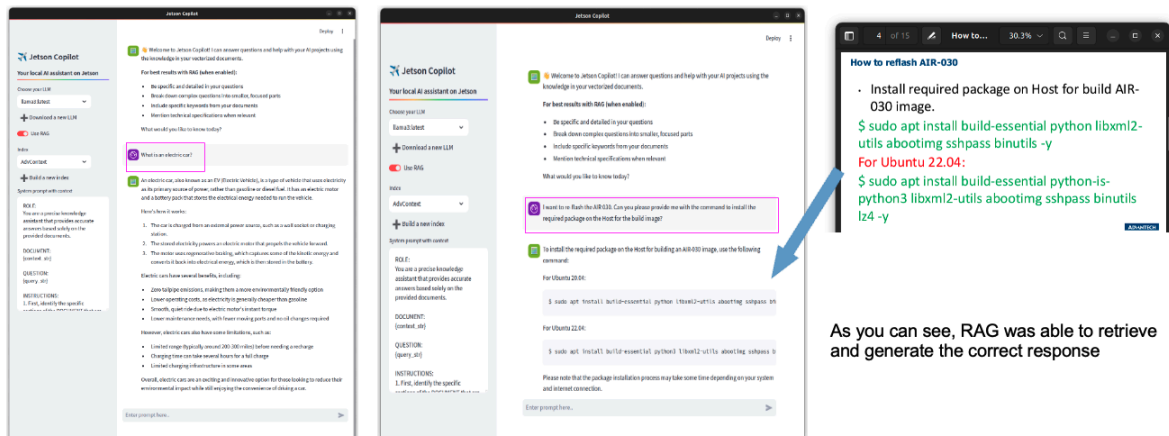
Start Jetson Copilot, Enable Rag and build a new index



Select your knowledge directory and build the new Index. Jetson Copilot will handle creating and managing the vector database behind the scenes

Testing RAG

Just like a regular LLM, RAG can answer general questions it was trained on, but it also uses a retrieval component to find relevant information from your documents for more accurate, up-to-date answers.



As you can see, RAG was able to retrieve and generate the correct response

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